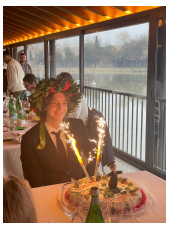
Filippo Pesce, M.Sc.

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https://github.com/SmoothManifold99



Profile

- Early-career physicist with a background in theoretical and mathematical physics, particularly quantum field theory, conformal field theories, and integrable models. Currently developing additional expertise in machine learning, with an interest in its applications to physics and financial markets.

Education

2022 – 2026 **M.Sc. Physics of the fundamental interactions, Università diPadova.**
Final grade: 110/110.
Thesis title: *Non-linear p-form gauge theories and their deformations.*
Thesis supervisors: Prof. Kurt Lechner, Prof. Dmitri Sorokin.
Employed a perturbative technique to study stress-tensor deformations of self-interacting chiral four-form theories in ten dimensions. Reviewed the PST and INZ formulations for the description of self-interacting chiral p-form theories and showed that, unlike in six dimensions, the solution of the root-$T\bar{T}$ flow equation in ten dimensions does not resum into a hyperbolic function of a Lorentz scalar function constructed from the auxiliary field.

2018 – 2022 **B.Sc. Physics, Università di Padova.**
Thesis title: *Poisson Structures and Hamiltonian Systems over the Formal Loop Space*.
Thesis supervisor: Prof. Paolo Rossi
Studied Poisson structures and Hamiltonian systems on the formal loop space, with particular attention to the KdV hierarchy.

Employment History

2025 – 2026 **Math teacher, high school.** Istituto Leonardo da Vinci, Portogruaro.
Taught mathematics to high-school students, adapting explanations of complex concepts to different levels of prior preparation.

Skills

Languages	Italian (native), English (professional proficiency).
Programming and Tools	Python, C++, LaTeX, Typst, Git, Mathematica
Machine Learning and Scientific Computing	scikit-learn, Pandas, Numpy, PyTorch, Keras.
AI-Assisted Development	OpenAI Codex, Claude Code.

References

Available on Request